

PRAFUL SAXENA

Senior Data Engineer | Real-Time Pipelines, AI-Driven Ops & Cloud-Scale Data Platforms

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PROFESSIONAL SUMMARY

Senior Data Engineer with 3+ years architecting and operating real-time, high-throughput data pipelines at One97 Communications (Paytm), consistently owning systems end-to-end — from design through production reliability. Maintained 99% uptime across streaming and transactional pipelines while cutting ETL costs 40% and job runtime 35% through performance tuning and resource optimization. Independently designed and shipped an autonomous AI-powered RCA platform (FastAPI, RAG, ChromaDB) that automates incident diagnosis end-to-end. Deep expertise in PySpark, Apache Kafka, and Airflow, with hands-on experience migrating legacy warehouses to Lakehouse architecture. Known for turning messy, high-volume data problems into reliable, self-serve systems that scale beyond a single engineer.

CORE SKILLS

Languages & Querying: Python, SQL, Scala

Big Data & Streaming: PySpark, Apache Kafka, Hadoop (MapReduce), Apache Airflow

Cloud & Data Platforms: AWS S3, Data Warehousing, Lakehouse Architecture, ETL/ELT Design, Data Modeling

ML / AI: Machine Learning, Deep Learning (CNN, RNN, LSTM), NLP, Prompt-Based AI Workflows

AI/LLM Systems: RAG, Vector Databases (ChromaDB), LLM Orchestration (Langflow), FastAPI, Prompt Engineering

Tools & Visualization: Tableau, Git/GitHub, Slack API

PROFESSIONAL EXPERIENCE

Manager, Data Strategy — SBI Card

Sep 2026 – Present

Recently joined; onboarding onto the team's data stack and priorities.

Senior Data Engineer — One97 Communications Ltd. (Paytm)

Mar 2023 – Sep 2026

- **Maintained 99% uptime** on production pipelines ingesting streaming and transactional data using PySpark, supporting both consumer and merchant tech features.
- **Cut ETL costs by 40% and job runtime by 35%** by re-architecting complex pipelines and optimizing resource allocation and performance tuning.
- **Architected and shipped an autonomous AI-powered Ops Monitoring & RCA platform** (FastAPI, ChromaDB, Langflow) — an 8-node pipeline that detects Spark/YARN/Azkaban failures every 60s, retrieves similar past incidents via RAG (reusing prior RCAs at zero LLM cost for repeat failures), and Slack-alerts the on-call engineer with root cause and fix — cutting mean-time-to-diagnosis from tens of minutes to a single automated alert.
- **Built a real-time customer location enrichment pipeline** on Kafka, replacing a 30-minute batch/SFTP process; streamed events end-to-end (consume → enrich → GeoHash derivation → sink topic → external adaptor → CleverTap update) to deliver live city/state updates to customer profiles.
- **Built a self-serve customer-to-phone-number mapping tool**, pairing a TSX dashboard with backend Scala jobs on Azkaban, letting business teams trigger mapping runs and receive secure, presigned-URL CSV exports via email — no engineering involvement required.
- **Led migration planning from a traditional data warehouse to Lakehouse architecture**, evaluating performance benchmarks, cost-optimization opportunities, and data governance for large-scale feature engineering pipelines.

Junior Engineer — NCC Limited

Apr 2018 – Mar 2023

- **Improved machinery and plant utilization** by performing statistical analysis of production and performance data, and categorizing assets to surface maximum-efficiency opportunities.
- **Managed relational databases (MySQL, SQL Server)** across multiple product lifecycle environments, ensuring data integrity, recovery, and space planning.
- **Built Hadoop/MapReduce jobs** (map-side and reduce-side joins) for large-scale data cleaning, pre-processing, and analytics.

PROJECTS

E-Commerce Shipping Delivery Prediction (Jun 2022 – Jul 2022)

- Built an EDA and preprocessing pipeline (NumPy, Pandas, PCA) to predict on-time delivery based on transport mode, location, and product availability.

Age, Gender & Ethnicity Detection via CNN (Oct 2022 – Nov 2022)

- Trained a CNN on the UTK real-time face dataset (OpenCV, Keras/TensorFlow) to predict age, gender, and ethnicity from image data.

EDUCATION, CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning — Imarticus Learning Institute, Lucknow (Dec 2022) · B.Tech, Mechanical Engineering — U.P.T.U, Lucknow (Jun 2014)
- Certifications: Advanced Python (Udemy) · Machine Learning (Internshala) · Six Sigma (IIBM)
- 7th rank, KPMG Hackathon (Skillenza) · Top 500, Kaggle Tabular Playground Series